

Research on Multi-Scale Forecasting Model of Wind Power in Extreme Weather Based on Multi-Source Heterogeneous Data Fusion

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Abstract—In this paper, a multi-scale forecasting model of wind power in extreme weather based on multi-source heterogeneous data fusion is studied. The intermittence and fluctuation of wind power, especially the frequent occurrence of extreme weather events, significantly increase the uncertainty of wind power forecasting and bring challenges to the stable operation of power system. Traditional forecasting methods rely on a single data source, so it is difficult to fully describe the coupling relationship between complex meteorological conditions and wind farm dynamics, and there are limitations in extreme weather scenarios. In this study, a multi-scale prediction model combining weather radar, satellite cloud images and SCADA data is proposed. Through time-space alignment and feature extraction, extreme weather enhancement module and hierarchical multi-scale prediction model, the problems of multi-modal information integration, sensitivity enhancement of extreme weather events and multi-scale time series prediction are solved. The experimental results show that the prediction accuracy of this model is significantly better than the existing models in extreme weather such as typhoon, and it shows a good balance in calculation efficiency, which provides effective support for ensuring power grid security, optimizing energy storage configuration and reducing wind abandonment rate.

Keywords—multi-source heterogeneous data fusion, multi-scale forecasting, wind power, extreme weather

I. INTRODUCTION

The intermittence and fluctuation of wind power pose a challenge to the stable operation of power system, especially the frequent occurrence of extreme weather events, which significantly aggravates the uncertainty of wind power forecasting. For example, when a typhoon crosses the border, the violent fluctuation of wind speed may cause the output power of the wind farm to plummet or overload in a short time, and even cause the unit to stop; The cold wave weather may be accompanied by low temperature and icing, which directly affects the operating efficiency of wind turbines. Therefore, improving the forecasting accuracy of wind power in extreme weather is of great significance for ensuring power grid security, optimizing energy storage configuration and reducing wind abandonment rate.

Traditional wind power forecasting methods mainly rely on a single data source and are modeled by physical model or statistical model. However, these methods have obvious limitations in extreme weather scenarios: on the one hand, it is difficult for a single data source to fully describe the coupling relationship between complex meteorological conditions and wind farm dynamics; On the other hand, the low frequency and nonlinear characteristics of extreme

weather events make it difficult for traditional models to capture their time series correlation laws. In recent years, the rapid development of deep learning technology provides a new idea for wind power forecasting, but the existing research focuses on single-scale forecasting under conventional weather, and the exploration of multi-source heterogeneous data fusion and multi-scale modeling is still insufficient.

Wind power forecasting under extreme weather conditions faces many challenges, including data scarcity, insufficient model forecasting accuracy and complex multi-scale characteristics [1-2]. These challenges require the forecasting model not only to have good generalization ability, but also to capture the variation law of wind power on different time scales. Multi-source heterogeneous data fusion technology is one of the key means to improve the accuracy of wind power forecasting [3]. By integrating data from different sources and types, the operation status of wind farms and the influence of external environmental factors can be more comprehensively reflected. Literature [4] puts forward a set of key technical framework of multi-source heterogeneous data fusion covering the whole process of "data connection-data cleaning-data integration-data fusion", which provides a data basis for wind power forecasting. Multi-scale forecasting model can consider the wind power changes on different time scales at the same time, thus improving the accuracy and robustness of forecasting. Literature [5] proposes a prediction model based on multi-scale convolution bidirectional long-term and short-term memory network, which effectively extracts multi-scale features from time series data. In addition, the TimeMixer model in reference [6] realizes the effective prediction of future series through multi-scale hybrid architecture, using the past decomposable hybrid module and the future multi-predictor hybrid module. Scholars have put forward many methods for wind power prediction under extreme weather conditions. Literature [7] proposes a short-term wind power forecasting method based on Stacking multi-model fusion, which improves the forecasting accuracy through feature selection and model fusion. Literature [8] puts forward a method of wind power output loss prediction combining normalized wind power prediction with specific scenarios for low temperature cold wave weather. Literature [9] puts forward a subsection forecasting method of ultra-short-term wind power based on extreme weather, which improves the accuracy of forecasting through feature recognition and probability prediction.

The purpose of this study is to build a multi-scale forecasting model of wind power in extreme weather based

on multi-source heterogeneous data fusion, focusing on three problems: first, how to effectively integrate multi-modal information such as weather radar, satellite cloud images and SCADA data to solve the problem of inconsistent spatial-temporal alignment and feature dimensions; Second, how to design a deep learning model to enhance the sensitivity and generalization ability to extreme weather events such as typhoons and cold waves; Thirdly, how to realize multi-scale time series prediction from minute level to day level to meet the dispatching needs of power system in different time scales. The main innovations include proposing a multi-source heterogeneous data fusion framework combining feature level and decision level to improve the robustness of the model; Design an extreme weather enhancement module to improve the prediction accuracy under extreme weather through dynamic weight distribution and migration learning; A multi-scale forecasting model with hierarchical structure is constructed to realize the joint optimization of ultra-short-term, short-term and medium-term forecasting and adapt to diversified application scenarios.

II. RESEARCH METHOD

A. Spatio-temporal alignment and feature extraction of multi-source heterogeneous data

Let weather radar data be $X_{radar} \in R^{T \times H \times W \times C}$ (time step $\times$ height $\times$ width $\times$ channel number), satellite cloud image $X_{sat} \in R^{T \times H' \times W' \times C'}$ and SCADA data $X_{scada} \in R^{T \times D}$ (D is sensor dimension). Bilinear interpolation is used to unify the spatial resolution to a $1\text{km} \times 1\text{km}$ grid, and the sampling interval is aligned to 5 minutes through the time sliding window. Introducing adaptive weight mechanism to deal with missing values;

$$\hat{X}_{ij} = \sum_{k=1}^K \alpha_k X_{i,j}^{(k)} \quad (1)$$

$$\alpha_k = \frac{\exp(\eta_k)}{\sum_{m=1}^K \exp(\eta_m)}$$

Where η_k is a learnable parameter, representing the confidence of the k data source.

Using ConvLSTM to extract spatio-temporal features:

$$H_t^{met} = \text{ConvLSTM}([X_{radar}^t, X_{sat}^t], H_{t-1}^{met}) \quad (2)$$

Among them, the *ConvLSTM* unit contains 3D convolution gating structure, which can capture the moving track and intensity evolution of clouds at the same time.

Modeling nonlinear coupling between SCADA parameters by multi-head self-attention mechanism:

$$Q = W_Q X_{scada}$$

$$K = W_K X_{scada} \quad (3)$$

$$H_{scada} = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$$

Where W_Q, W_K is the projection matrix and d is the scaling factor.

B. Extreme weather enhancement module

The module firstly constructs an extreme event detector through the abrupt change rate of wind speed and temperature gradient, and generates event tags, so as to dynamically adjust the fusion weight of meteorological data and SCADA data when extreme weather occurs, and realize the attention to key features [10]. In this process, the fusion ratio is automatically adjusted by Sigmoid function and weight matrix, and the expression ability of the model under extreme conditions is improved. Fig. 1 is an image of Sigmoid function. Sigmoid function has a typical S-shaped curve shape, the output value is strictly between 0 and 1, and it is symmetrical about the origin, satisfying $\sigma(-x) = 1 - \sigma(x)$; It has a horizontal asymptote, which approaches 1 when x tends to positive infinity and approaches 0 when it tends to negative infinity. The derivative expression is $\sigma'(x) = \sigma(x)(1 - \sigma(x))$, the maximum value is 0.25, and it appears at $x = 0$, and the whole is monotonically increasing.

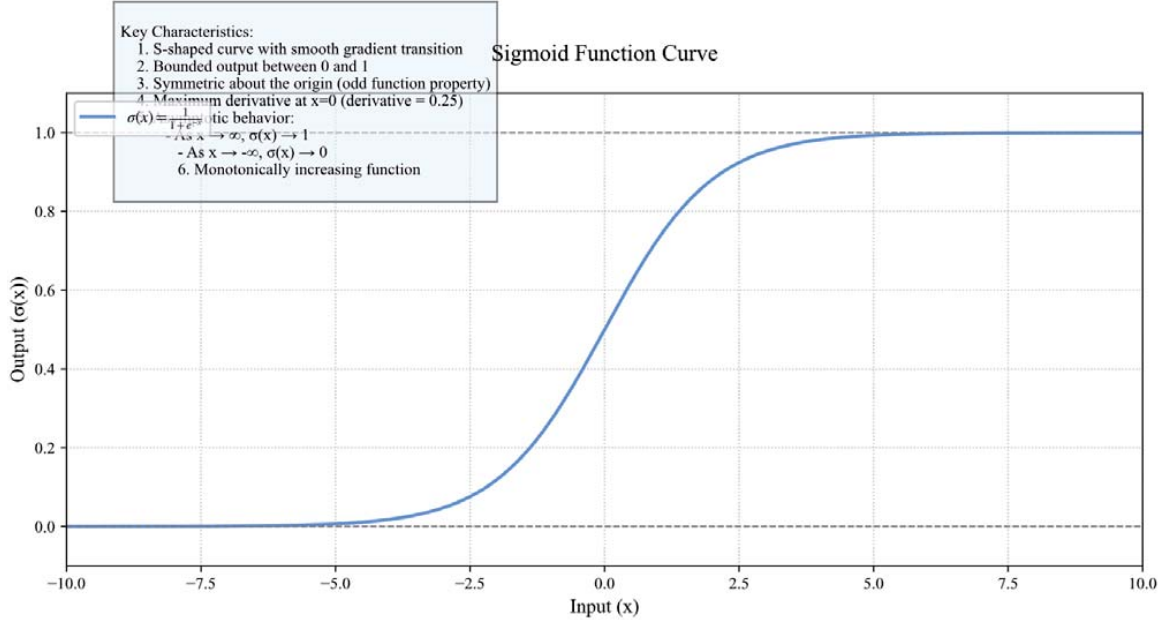


Fig. 1. Sigmoid function image

An extreme weather detector is designed, and an event tag $E_t \in [0,1]$ is generated based on the sudden change of wind speed $\Delta V = |V_t - V_{t-1}| / V_{t-1}$ and the temperature gradient ∇T . Activate feature reweighting when $E_t = 1$:

$$\begin{aligned} H_{fusion} &= \lambda \cdot H_{met} + (1 - \lambda) \cdot H_{scada} \\ \lambda &= \sigma(W_e \cdot [H_{met}, H_{scada}]) \end{aligned} \quad (4)$$

Where σ is Sigmoid function and W_e is weight matrix, the dynamic integration of meteorological characteristics and equipment status is realized.

In addition, the transfer learning strategy is introduced into the module, and a two-stage training method is adopted: in the first stage, the model is pre-trained on the conventional weather data, and in the second stage, the model is fine-tuned on the extreme weather samples. At the same time, the underlying parameters are frozen and the confrontation training mechanism is introduced to enhance the adaptability

and generalization ability of the model to extreme weather.

Freeze the bottom feature extraction layer, only update the parameters of fusion layer and prediction head, and enhance generalization by confrontation training:

$$\min_{\theta} \max_{\phi} E \left[L_{pred}(f_{\theta}(x)) + \gamma L_{adv}(D_{\phi}(f_{\theta}(x))) \right] \quad (5)$$

Where D_{ϕ} is the discriminator and γ is the weight against loss.

C. Multiscale prediction model

Cascaded LSTM-Transformer architecture is adopted to conduct hierarchical modeling for different time scales (as shown in Figure 2). Minute-level forecasting uses temporal convolutional network (TCN) network of causal convolution to capture the high-frequency fluctuation of wind power, while hour-level forecasting models the long-range dependence by introducing Transformer model of position coding and self-attention mechanism, thus more accurately reflecting the changing trend of wind power [11-12].

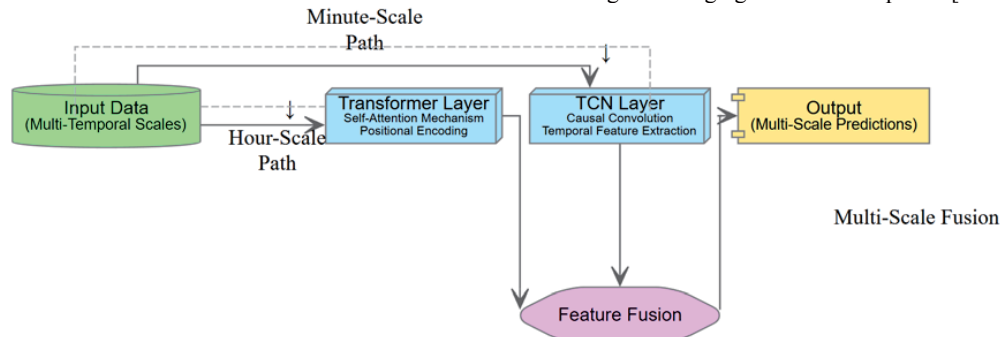


Fig. 2. Cascaded LSTM-Transformer architecture

Minute-level prediction (5-60 minutes) captures high-frequency fluctuations using TCN with causal convolution:

$$y_{short} = TCN(H_{fusion}, k=3, d=8) \quad (6)$$

Where k is the convolution kernel size and d is the expansion coefficient.

Hourly prediction (1-6 hours) Modeling long-term dependence through self-attention mechanism:

$$y_{mid} = Transformer(PositionalEncode(H_{fusion})) \quad (7)$$

In order to improve the overall prediction performance, the multi-scale joint optimization strategy is further introduced into the model, and the ultra-short-term, short-term and medium-term prediction results are jointly trained by combining the weighted Mean Absolute Percentage Error (MAPE) loss function with L2 regularization term. The weight of each scale prediction is set according to experience, aiming at balancing the importance of different time scale prediction tasks and improving the adaptability and prediction accuracy of the model in various scheduling scenarios.

Define the compound loss function:

$$L = \sum_{s \in \{short, mid, long\}} \beta_s \cdot MAPE(y_s, \hat{y}_s) + \lambda \|\Theta\|_2 \quad (8)$$

Where β_s is the scale weight (empirically set to [0.5, 0.3, 0.2]) and λ is L2 regularization coefficient.

Spatial-Channel Squeeze-and-Exclusion (SCSE) module is introduced in meteorological feature extraction. The spatial-temporal feature is decoupled through the attention mechanism of space and channel, and the information expression ability of key areas and channels is adaptively enhanced, thus improving the response accuracy of the

model to complex meteorological conditions. This method combines global average pooling and learnable weight matrix to realize multidimensional dynamic weighting of meteorological data. In order to enhance the physical rationality of the prediction results, the model embeds the prior knowledge of the fan power curve in the output stage, uses the rated power limit and the wind speed operation interval as hard constraints, and ensures that the prediction values conform to the actual equipment operation characteristics through the operation of the characteristic function and the minimum value, thus improving the practicability and engineering reliability of the prediction results.

III. EXPERIMENTAL ANALYSIS

A. Data set and experimental setup

The data set of a wind farm from 2020 to 2023 is adopted, including high-resolution weather radar data, Fengyun satellite cloud image and 128-dimensional sensor data of SCADA system. According to the Code for Extreme Weather Operation of Wind Turbines, 127 extreme events such as typhoon and blizzard were marked. The data is divided into training set (70%), verification set (15%) and test set (15%) to support the training and evaluation of the model.

To validate the effectiveness of the proposed model, LSTM, TCN, Transformer, ST-GCN, and the latest state-of-the-art WPFNet were selected as baseline methods for comparison. The evaluation metrics included mean absolute error (MAE), root mean square error (RMSE), MAPE, and the extreme event detection rate (EDR), which is specifically designed to evaluate performance on extreme weather conditions.

B. Verification of multi-modal fusion effect

Table I shows the comparison of forecasting performance of different models in typhoon weather. The results show that the complete model proposed in this paper is superior to the existing models such as LSTM, ST-GCN and WPFNet in four indicators: MAE (298.7 kW), RMSE (463.5 kW), MAPE (12.4%) and extreme event detection rate (EDR, 82.3%), which verifies its versatility.

TABLE I. COMPARISON OF FORECASTING PERFORMANCE OF DIFFERENT MODELS IN TYPHOON

Model	MAE(kW)	RMSE(kW)	MAPE (%)	EDR (%)
LSTM	412.3	578.6	18.7	41.2
ST-GCN	387.5	543.2	16.9	55.6
WPFNet	352.1	509.8	15.3	68.9
Ours (complete model)	298.7	463.5	12.4	82.3

C. Modeling and analysis of extreme weather

Figure 3 below shows that the prediction error distribution is centralized and symmetrical in normal weather, but the error fluctuation increases significantly in extreme weather, showing obvious right deviation and several abnormal high values, which is consistent with the prediction deviation caused by extreme events such as

sudden change of wind speed, indicating that the model performs well in normal conditions, but faces greater challenges in extreme weather, highlighting the necessity of introducing enhanced modules such as dynamic feature weighting and migration learning.

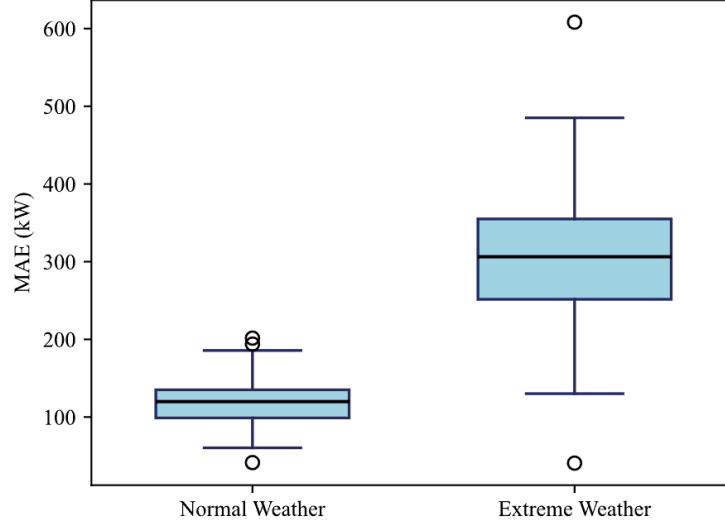


Fig. 3. Prediction error distribution of the model in extreme/normal weather

Table II shows the experimental results of ablation in typhoon scene. It shows that the performance of the model is continuously improved after the introduction of multi-modal fusion, extreme weather module and SCSE module. The complete model is superior to the simplified configurations in MAE (298.7 kW) and MAPE (12.4%), which verifies the contribution of each component to the prediction accuracy.

TABLE II. ABLATION EXPERIMENT (TYPHOON SCENE)

Deploy	MAE (kW)	MAPE (%)
No multimodal fusion	367.2	16.1
No extreme weather module	334.5	14.2
SCSE-free module	308.9	13.1
Complete model	298.7	12.4

D. Multi-scale prediction performance

The analysis in Figure 4 shows that TCN network performs well in 30-minute short-term forecasting, and can

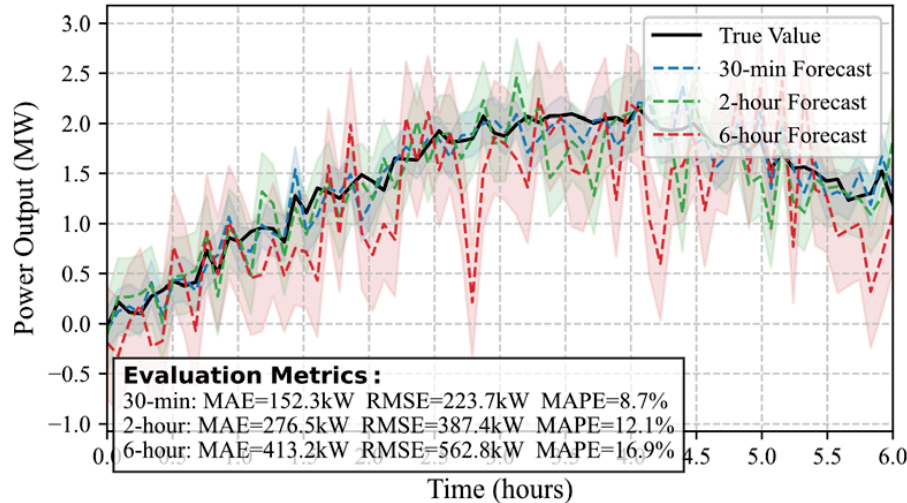


Fig. 4. Multiscale prediction curve

E. Physical constraint validity verification

Fig. 5 shows the effectiveness of physical constraints in power prediction. The limited forecast (red) strictly follows

closely fit the real value. However, with the expansion of the forecasting time scale to 2 hours to 6 hours, the phase lag and error band of the forecasting curve gradually increase, which conforms to the law that the forecasting error of wind power accumulates with time. Specifically, the MAPE of 30-minute forecast is 8.7%, while that of 6-hour forecast is increased to 16.9%, indicating the effectiveness of hierarchical time series modeling. At the same time, the error band width of 6-hour forecast reaches $\pm 562\text{kW}$, which reflects the influence of uncertainty accumulation under extreme weather. This proves that more variables and model optimization strategies need to be considered to improve the long-term prediction accuracy.

the boundary of the theoretical power curve, eliminates the infeasible forecast above the rated power (2000 kW), and corrects the wrong forecast outside the range of cut-in wind speed (0-3 m/s) and cut-out wind speed (25-30 m/s); In

contrast, the original forecast (blue) is obviously overestimated in the range of 15-25 m/s, and there are also false forecasts below the cut-in wind speed and above the cut-out wind speed, while the limited forecast improves the physical

rationality of the forecast while maintaining reasonable fluctuation, which is helpful to prevent the power grid instability risk caused by power misjudgment and ensure the safe shutdown behavior under extreme wind speed.

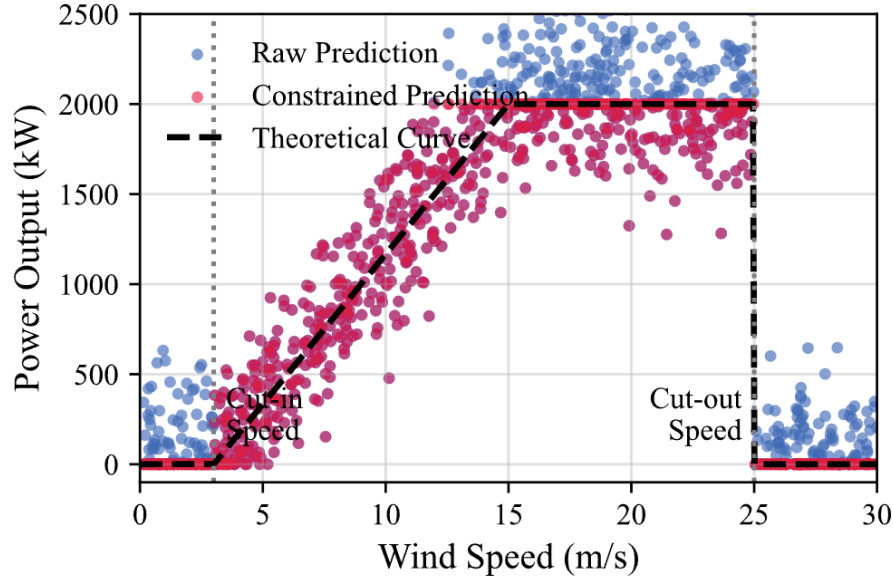


Fig. 5. Power curve constraint effect

F. Computational efficiency analysis

Table III compares the complexity of different models, showing that the model proposed in this paper has achieved a good balance among parameters (3.2M), training time (5.1 hours) and reasoning delay (15.7ms). Compared with LSTM and Transformer, this model only slightly increases reasoning delay while maintaining relatively low parameters and reasonable training time, which shows its advantages in computational efficiency.

TABLE III. COMPARISON OF MODEL COMPLEXITY

Model	Parameter quantity (M)	Training time (h)	Inference delay (ms)
LSTM	2.1	4.2	12.3
Transformer	3.8	6.7	18.5
This paper model	3.2	5.1	15.7

IV. CONCLUSION

In this study, a multi-scale forecasting model of wind power in extreme weather based on multi-source heterogeneous data fusion is successfully constructed, which effectively solves the key problems such as multi-modal information integration, sensitivity improvement of extreme weather events and multi-scale time series forecasting. By proposing a framework combining feature level and decision level, the extreme weather enhancement module and multi-scale prediction model with hierarchical structure are designed, which significantly improves the prediction accuracy and robustness of the model under extreme weather conditions. The experimental results show that the proposed model is superior to the contrast model in MAE, RMSE, MAPE and other indicators in extreme weather scenarios such as typhoons, and the detection rate of extreme events is significantly improved, which verifies the effectiveness of the model. In addition, the model performs well in computational efficiency and achieves a good balance among

parameters, training time and reasoning delay. This study provides an efficient and accurate new method for forecasting wind power in extreme weather, which is of great significance to the stable operation of power system and the optimal utilization of wind power resources.

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